HORIZONTAL COMPOSITION COMPATIBILITY OF THE EXCHANGE MAPS FOR A SIDE-BY-SIDE PAIR OF PROPER-AND-OPEN SQUARES

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Allowed standard prerequisites (editorial).

- Exchange map for proper and separated locally quasi-finite morphisms (Lemma lemma-shriek-proper-and-open, canonical tag 0F7B). cited at statement line 2907 and proof lines 2934-2941. Supplies the two square exchange maps; its proof shows their representing maps are quasi-isomorphisms.
- Exactness of lower shriek along separated locally quasi-finite morphisms (Lemma lemma-lqf-f-shriek-separated-colimits, canonical tag 0F57). cited at proof lines 2942-2945. Exactness of g_! upgrades the composite quasi-isomorphism to the one representing the rectangle's map.
- Composition of lower shriek along separated morphisms (Lemma lemma-f-shriek-composition, canonical tag 0F50). cited at proof lines 2913-2916. The equalities (h after h')_! = h_! h'_! and (g after g')_! = g_! g'_! used to spell out the statement.

Source reference. Original lemma 0F50: sources/stacks-chow/tags.txt, source lines 18920–18920. Cited at proof lines 2913-2916: the equalities (h after h')_! = h_! h'_! and (g after g')_! = g_! g'_! used to spell out the statement.

Source reference. Original lemma 0F57: sources/stacks-chow/tags.txt, source lines 18927–18927. Cited at proof lines 2942-2945: exactness of g_! makes the composite quasi-isomorphism represent the rectangle's map.

Source reference. Original lemma 0F7B: sources/stacks-chow/tags.txt, source lines 19001–19001. Cited at statement line 2907 and proof lines 2934-2941: supplies the two square exchange maps and the proof that their representing maps are quasi-isomorphisms.

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Lemma 0.2. *Consider a commutative diagram of schemes*

$$\begin{array}{ccccc} X'' & \xrightarrow[g']{} & X' & \xrightarrow[g]{} & X \\ {\scriptstyle f''}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y'' & \xrightarrow{h'} & Y' & \xrightarrow{h} & Y \end{array}$$

with f, f', and f'' proper and g, g', h, and h' separated and locally quasi-finite. Then the isomorphisms of Lemma 0F7B for the two squares compose to give the isomorphism for the outer rectangle (see proof for a precise statement).

Proof. The statement means that if we write $(h \circ h')_! = h_! \circ h'_!$ and $(g \circ g')_! = g_! \circ g'_!$ using the equalities of Lemma 0F50, then the isomorphism $h_! \circ h'_! \circ Rf''_* \to Rf_* \circ g_! \circ g'_!$ of the outer rectangle is equal to the composition

$$h_! \circ h'_! \circ Rf''_* \to h_! \circ Rf'_* \circ g'_! \to Rf_* \circ g_! \circ g'_!$$

of the two maps of the squares in the diagram. To prove this choose a K-injective complex $\mathcal{I}^\bullet$ of Λ-modules on $X''_{étale}$ and a quasi-isomorphism $g'_!\mathcal{I}^\bullet \to \mathcal{J}^\bullet$ to a K-injective complex $\mathcal{J}^\bullet$ of Λ-modules on $X'_{étale}$. Next, choose a quasi-isomorphism $g_!\mathcal{J}^\bullet \to \mathcal{K}^\bullet$ to a K-injective complex $\mathcal{K}^\bullet$ of Λ-modules on $X_{étale}$. The proof of Lemma 0F7B shows that the canonical maps

$$h'_! f''_* \mathcal{I}^\bullet \to f'_* \mathcal{J}^\bullet \quad \text{and} \quad h_! f'_* \mathcal{J}^\bullet \to f_* \mathcal{K}^\bullet$$

are quasi-isomorphisms and these quasi-isomorphisms define the first and second arrow above. Since $g_!$ is an exact functor (Lemma 0F57) we find that $g_! g'_! \mathcal{I}^\bullet \to \mathcal{K}^\bullet$ is a quasi-ismorphism and hence the canonical map

$$h_! h'_! f''_* \mathcal{I}^\bullet \to f_* \mathcal{K}^\bullet$$

is a quasi-isomorphism and represents the map for the outer rectangle in the derived category. Clearly this map is the composition of the other two and the proof is complete. □